

Project Dataset Description: ESN Porto Student Registrations

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1 Dataset Overview and Time Span

Our dataset contains activation records of the **ESNcard**, an annual membership acquired by international students upon arriving in Porto to access ESN (Erasmus Student Network) events and partner discounts (housing, transportation, and lifestyle) across 42 countries. Each entry includes the registration timestamp and basic demographic and academic information.

The dataset covers **September 2020 to November 2025**, totaling approximately 65 months across six academic years. After cleaning (removing invalid ages and date anomalies), the dataset contains **14,526 registrations**.

2 Variables Observed

- **Temporal:** *Registration Date* (Timestamp).
- **Demographic:** *Age* (derived from birthdate), *Gender*, and *Country* of origin.
- **Academic:** *University* (e.g., U.Porto, IPP), *Program Type* (Erasmus, Exchange, Volunteer), and *Program Duration*.

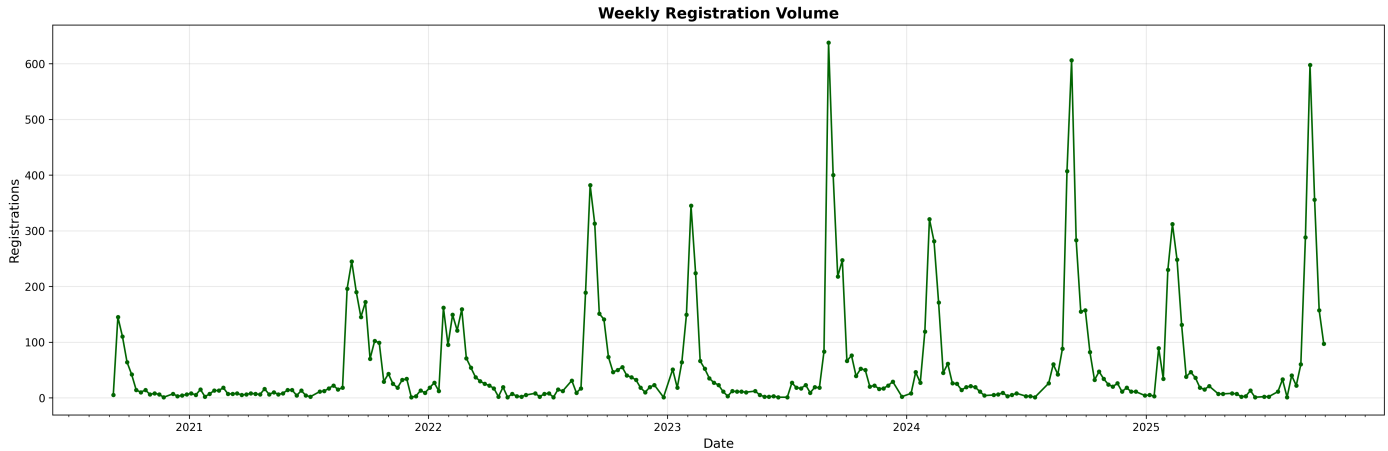


Fig. 1. Weekly Registration Volume from late 2020 to 2025

3 Frequency Selection: Weekly

Since the raw data is recorded as precise timestamps, we have the flexibility to aggregate the time series at any resolution. Daily frequency proved problematic, with most days recording zero registrations interrupted by extreme spikes during semester starts, creating excessive noise and sparsity. Monthly aggregation provided the cleanest seasonal signal but sacrificed temporal detail. We choose weekly frequency as it smoothed the daily volatility while preserving the semester-based arrival patterns.

4 Data Characteristics and Limitations

The resulting weekly time series has some seasonality peaking in **September** (Fall semester) and **February** (Spring semester). There is an underlying upward trend.

The dataset faces two primary limitations:

1. **Incomplete Current Year:** Data for the 2025-2026 academic year is ongoing. This creates misleading apparent declines in the final observations and hinders reliable year-over-year comparisons for the current cycle.
2. **Extreme Seasonality:** Registrations are highly concentrated in specific intake weeks, leaving extended periods (mid-semester and summer) of minimal activity, which reduces the effective sample size for analyzing off-peak dynamics.